Introduction to Cybersecurity

Cybersecurity Hands-On Lab

Module 1: Understanding Basic Security Weaknesses

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1. Part 1: Password Security

1.1 Analysis of Weak Passwords

The following passwords were examined: 123456, password, admin, 12345678, qwerty. These passwords are highly insecure because they are very short, contain no complexity, and are among the most commonly used worldwide. An automated brute force or dictionary attack tool can crack them in under a second.

1.2 Strong Passwords Created

Three strong passwords were created following the required rules (minimum 12 characters, uppercase and lowercase, numbers, special characters):

1. **N@d1nga_S3cur3!2026**
2. **Ph@r3s#Cyber!W0rk**
3. **Y!3n0uy@ba_S3cure$Lab**

2. Part 2: Observing Network Traffic

Wireshark was launched and the active network interface was selected. Packet capture was started by double-clicking the interface. Several websites were then visited (Google, Wikipedia, university website) for approximately two minutes to generate traffic. The capture was then stopped using the red Stop button.

3. Part 3: Packet Analysis

3.1 DNS Traffic

The filter dns was applied to isolate DNS queries. Domain name resolution requests were visible, showing how each website address is translated into an IP address before any connection is made. One domain observed: www.bing.com.

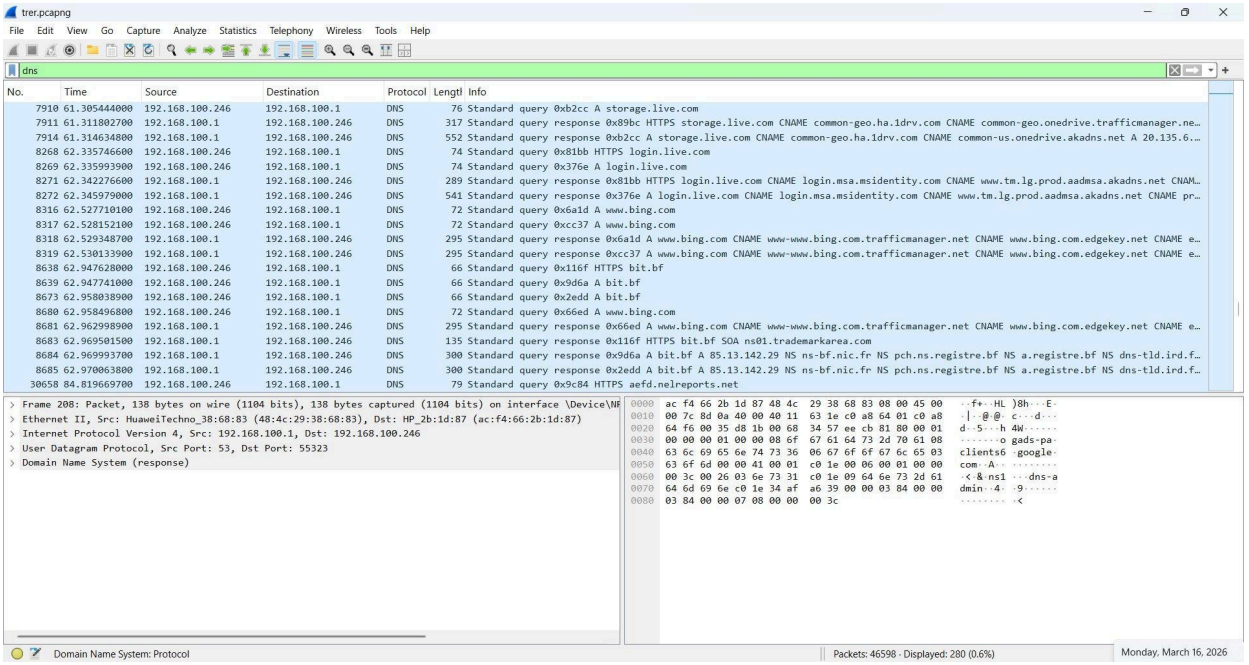

Figure 1: Wireshark – DNS filter showing domain name resolution queries

3.2 HTTP Traffic

The filter http was applied to observe unencrypted web traffic. The captured packets revealed request URLs and HTTP headers in plain text, demonstrating that this type of traffic is fully readable by anyone intercepting the connection.

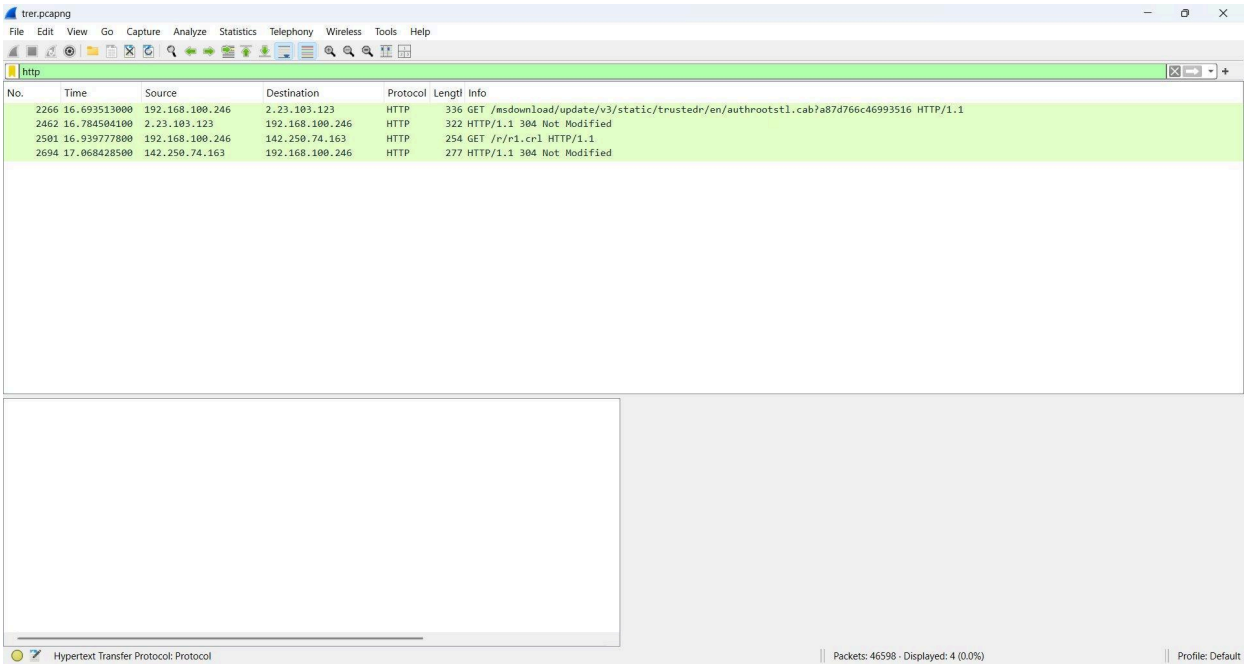

Figure 2: Wireshark – HTTP filter showing unencrypted request data

3.3 Encrypted TLS Traffic

The filter tls was applied to isolate encrypted HTTPS communications. Unlike HTTP, the content of TLS packets is unreadable without the decryption keys, confirming that encryption effectively protects data in transit.

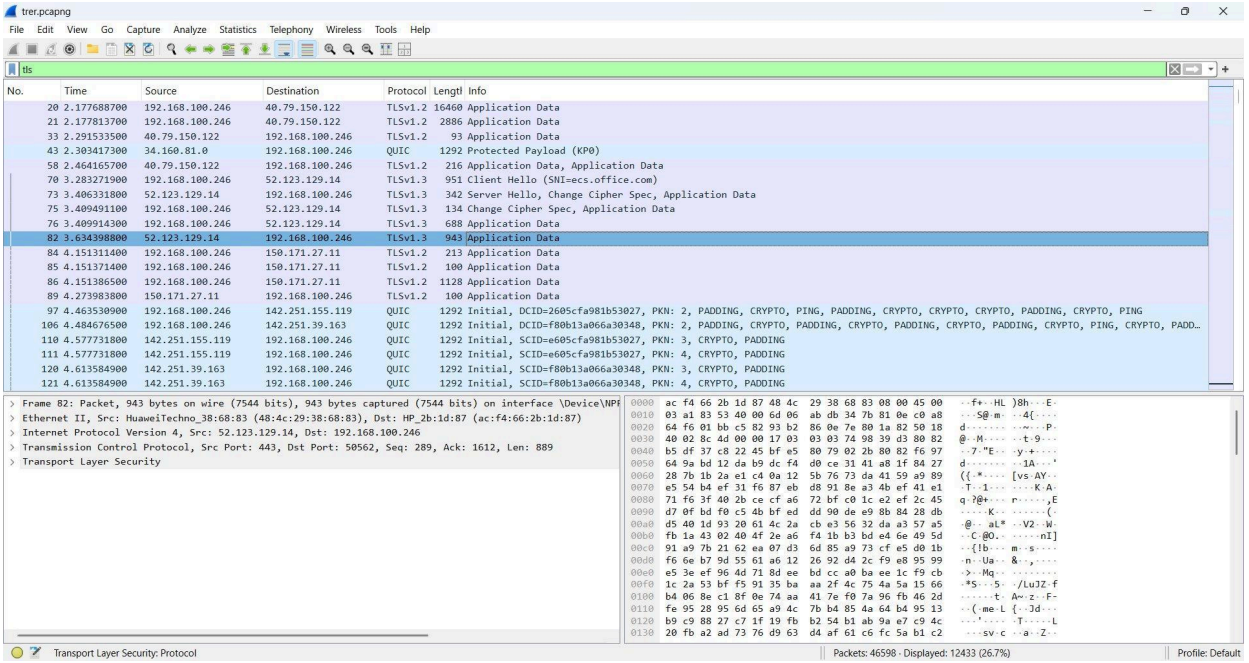

Figure 3: Wireshark – TLS filter showing encrypted application data

4. Analysis Questions

1. *Approximately how many packets were captured?*

A total of 46,598 packets were captured during the session, as indicated in the Wireshark status bar.

2. *List three protocols observed during the experiment.*

Three protocols observed: DNS (domain name resolution), HTTP (unencrypted web transfers), and TLS/TLSv1.2/TLSv1.3 (encrypted HTTPS communications).

3. *Why is encryption important when browsing the internet?*

Encryption ensures that data exchanged between a user and a server cannot be read by third parties. Without it, sensitive information such as passwords, personal data, or session tokens travels in plain text and can be intercepted by anyone on the network.

4. *What type of information could attackers obtain from unencrypted traffic?*

From unencrypted HTTP traffic, an attacker could obtain visited URLs, form data, HTTP headers, session cookies, and authentication credentials. This clearly shows why HTTPS must be used on any site that handles sensitive data.